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mshqc: A C++20 Quantum Chemistry Engine for Orbital-Optimized Møller–Plesset Perturbation Theory

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Software

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Summary

mshqc is a high-performance quantum chemistry library implementing wave-function-based electronic structure methods in a compact C++20 core with a Python interface built on nanobind (Jakob, 2022). It provides restricted, unrestricted, and restricted open-shell Hartree–Fock references (RHF, UHF, ROHF) (Pople & Nesbet, 1954; Roothaan, 1951, 1960) with Superposition of Atomic Densities (SAD) initial guesses and Pulay's DIIS extrapolation (Pulay, 1980, 1982), alongside canonical and semicanonical (RMP2, UMP2, ROMP2, RMP3, UMP3, ROMP3) (Head-Gordon et al., 1988; Møller & Plesset, 1934; Pople et al., 1976) and orbital-optimized (OMP2, OMP3) (Bozkaya et al., 2011; Bozkaya, 2011; Neese et al., 2009) Møller–Plesset perturbation theory.

To overcome four-center electron repulsion integral (ERI) bottlenecks evaluated via libcint (Sun, 2015), mshqc unifies three integral modes across all solvers: exact four-center tensors (exact) with Schwarz screening (Häser & Ahlrichs, 1989), full-pivoted Cholesky decomposition (cholesky) (Beebe & Linderberg, 1977; Koch et al., 2003), and auxiliary-basis Density Fitting (df) (Feyereisen et al., 1993; Weigend, 2002). High-dimensional contractions are dispatched to TBLIS (Matthews, 2017) and BLAS Level-3 kernels, while orbital rotations in OMP2 and OMP3 are driven by a matrix-free Steihaug–Toint trust-region Second-Order SCF (TrustRegionSOSCF) optimizer.

Statement of need

Orbital-optimized Møller–Plesset methods (OMP2, OMP3) relax reference orbitals in the presence of electron correlation, restoring Hellmann–Feynman satisfaction for one-electron properties and eliminating artificial spin contamination in open-shell radicals (Bozkaya et al., 2011; Bozkaya, 2011; Neese et al., 2009). However, each macro-iteration requires re-transforming two-electron integrals and evaluating $\mathcal{O}(N^5)$ (OMP2) or $\mathcal{O}(N^6)$ (OMP3) contractions, making naive implementations memory- and compute-bound.

Researchers and students prototyping orbital-optimization algorithms or three-index tensor factorizations face a steep barrier to entry in large production suites due to deep abstraction layers. mshqc provides an inspectable C++20 codebase where SCF solvers, cholesky/df integral transformers, TBLIS contractions, and trust-region SOSCF optimizers are unified under a two-tier API: users can run multi-method pipelines in one line via mshqc.run() or instantiate C++20 classes directly in Python to inspect intermediate tensors.

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State of the field

Production packages such as `Psi4` (Smith et al., 2020) (`fnocc`/`dfocc`) and `PySCF` (Sun et al., 2020) provide comprehensive canonical and orbital-optimized perturbation theory, but modifying their internal integral transformation pipelines or Hessian preconditioners requires navigating hundreds of thousands of lines of code. Conversely, lightweight educational codes typically implement only canonical `RHF` and `RMP2` in pure Python, lacking open-shell references (`UHF`/`ROHF`), third-order correlation (`MP3`/`OMP3`), point-group symmetry, or three-index integrals. `mshqc` bridges this gap with a compact C++20 architecture that matches reference energies to machine precision while achieving multi-fold speedups on `OMP2` and `OMP3` workloads.

Software design

`mshqc` is structured into four C++20 layers exposed to Python via `nanobind` (Jakob, 2022) with zero-copy `Eigen` (Guennebaud et al., 2010) interoperability and automatic `GIL` release:

1. **Symmetry & Basis Infrastructure (`Molecule`, `BasisSet`, `PointGroup`, `PetiteList`):** Parses 538 bundled Gaussian basis sets (`.gbs`), including Dunning `cc-pVXZ` (Dunning, 1989) and `-RI`/`-JKFIT` auxiliary bases (Weigend, 2002), with contraction normalization (Helgaker et al., 2000), canonical axis alignment, and `PetiteList` symmetry reduction (Atkins et al., 1970).
2. **Unified Integral Engine (`IntegralEngine`, `Screening`, `CholeskyERI`, `DensityFittingERI`):** Evaluates integrals via `libcint` (Sun, 2015) under a unified `eri_method` (`"exact"`, `"cholesky"`, `"df"`), reducing storage from $\mathcal{O}(N^4)$ to $\mathcal{O}(N_{\text{aux}}N^2)$ three-center tensors B_{pq}^{Q} with optional `HDF5` out-of-core storage (The HDF Group, 1997). `CholeskyERI` employs unrestricted global diagonal pivoting across one- and two-center shell pairs with an unamplified Schwarz screening cutoff ($\tau_{\text{screen}} = \min(10^{-12}, 10^{-4}\tau_{\text{CD}})$) to prevent $1/\sqrt{D_{\max}}$ error amplification.
3. **Self-Consistent Field (`RHF`, `UHF`, `ROHF`):** Shares a common `BaseSCF` architecture with `SAD` guesses, lock-free `incore` quartet caching, `DIIS` extrapolation, spin-resolved Frobenius density convergence ($\Delta P = \sqrt{\|\Delta\mathbf{P}^{\alpha}\|_F^2 + \|\Delta\mathbf{P}^{\beta}\|_F^2}$), and post-convergence canonical Fock synchronization without `DIIS` mixing.
4. **Correlated & Orbital-Optimized Solvers (`RMP2`, `UMP2`, `ROMP2`, `OMP2`, `RMP3`, `UMP3`, `ROMP3`, `OMP3`):** Dispatches four-index contractions to `TBLIS` (Matthews, 2017) and three-index Z-vector response contractions to `OpenMP` + `BLAS` Level-3 kernels (Dagum & Menon, 1998), coupled with `TrustRegionSOSCF` orbital optimization.

Validation and performance

All solvers in `mshqc` were validated against `Psi4` (Smith et al., 2020) across closed-shell (`Ne`, H_2O) and open-shell (`Li`, `C`, `O`) systems in the `cc-pVTZ` (`cc-pVTZ-RI`) basis at a 10^{-9} Ha convergence and Cholesky threshold. Across all 75 benchmark runs (`5 systems × 3 ERI modes × 5 methods`: `RHF`/`UHF`, `RMP2`/`UMP2`, `RMP3`/`UMP3`, `OMP2`, `OMP3`), `mshqc` reproduces `Psi4` reference energies with zero deviation ($|\Delta E| = 0.00$ μHa across all 75/75 runs at 10^{-8} Ha precision).

As shown in Figure 1 and Table 1, `mshqc` achieves wall-clock speedups of $3.27\times$ to $97.25\times$ on `OMP2` and $5.06\times$ to $81.11\times$ on `OMP3` over `Psi4` (and up to $20.32\times$ on `MP3`), converging the orbital gradient norm $\|\mathbf{g}\|_2 < 10^{-5}$ Ha in 3 to 4 macro-iterations. On H_2O ($N = 58$, $N_{\text{aux}} = 141$), three-center factorizations reduce ERI memory by $23.9\times$ (90.5 MB to 3.8 MB) and lower `OMP3` wall-clock time from 0.628 s (`exact`) to 0.250 s (`df`).

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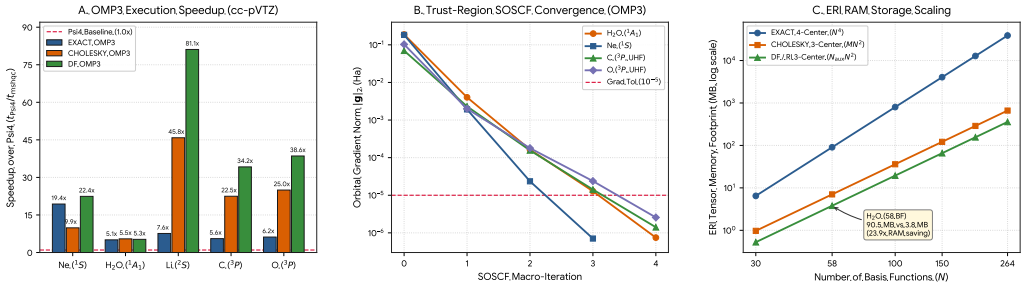


Figure 1: Benchmark evaluation of `mshqc` vs `Psi4` in the `cc-pVTZ` (`cc-pVTZ-RI`) basis: (A) `OMP3` wall-clock speedup ($t_{\text{Psi4}}/t_{\text{mshqc}}$) across `exact`, `cholesky`, and `df` modes; (B) Trust-Region SOSCF orbital gradient norm $\|\mathbf{g}\|_2$ convergence per macro-iteration; (C) ERI tensor RAM scaling comparing four-center `exact` ($\mathcal{O}(N^4)$) against three-center `cholesky` ($\mathcal{O}(MN^2)$) and `df` ($\mathcal{O}(N_{\text{aux}}N^2)$).

Table 1: Representative benchmark comparison between `mshqc` and `Psi4` (`cc-pVTZ` / `cc-pVTZ-RI`, threshold 10^{-9} Ha), reporting total energy, absolute deviation $|\Delta E|$ (μHa), wall-clock times (s), and speedup.

System	ERI	Method	Energy (Ha)	$\lvert\Delta E\rvert$	`mshqc` (s)	`Psi4` (s)	Speedup
Ne (^1S)	exact	RHF	-128.53186164	0.00	0.009	0.523	58.11x
Ne (^1S)	exact	RMP3	-128.80972987	0.00	0.019	0.386	20.32x
Ne (^1S)	exact	OMP2	-128.81022310	0.00	0.019	0.379	19.95x
Ne (^1S)	exact	OMP3	-128.81027068	0.00	0.038	0.736	19.37x
Ne (^1S)	cholesky	OMP3	-128.81027068	0.00	0.074	0.730	9.86x
Ne (^1S)	df	OMP2	-128.81022440	0.00	0.006	0.302	50.33x
Ne (^1S)	df	OMP3	-128.81042564	0.00	0.023	0.516	22.43x
H_2O ($^1A_1$)	exact	RHF	-76.05712742	0.00	0.101	0.574	5.68x
H_2O ($^1A_1$)	exact	OMP2	-76.33406054	0.00	0.354	1.158	3.27x
H_2O ($^1A_1$)	exact	OMP3	-76.33711056	0.00	0.628	3.180	5.06x
H_2O ($^1A_1$)	cholesky	OMP2	-76.33406054	0.00	0.256	1.588	6.20x
H_2O ($^1A_1$)	cholesky	OMP3	-76.33711056	0.00	0.831	4.537	5.46x
H_2O ($^1A_1$)	df	RMP2	-76.33220605	0.00	0.002	0.052	26.00x
H_2O ($^1A_1$)	df	OMP2	-76.33402331	0.00	0.057	0.470	8.25x
H_2O ($^1A_1$)	df	OMP3	-76.33725847	0.00	0.250	1.315	5.26x
Li (^2S, UHF)	exact	OMP2	-7.44391221	0.00	0.014	0.459	32.79x

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System	ERI	Method	Energy (Ha)	$\lvert\Delta E\rvert$	mshqc (s)	Psi4 (s)	Speedup
Li (^2S, UHF)	cholesky	OMP3	-7.44569499	0.00	0.037	1.696	45.84x
Li (^2S, UHF)	df	OMP2	-7.44391392	0.00	0.008	0.778	97.25x
Li (^2S, UHF)	df	OMP3	-7.44569708	0.00	0.019	1.541	81.11x
C (^3P, UHF)	exact	OMP3	-37.78269153	0.00	0.194	1.076	5.55x
C (^3P, UHF)	cholesky	OMP3	-37.78269153	0.00	0.093	2.091	22.48x
C (^3P, UHF)	df	OMP2	-37.76763043	0.00	0.013	0.713	54.85x
C (^3P, UHF)	df	OMP3	-37.78281240	0.00	0.045	1.539	34.20x
O (^3P, UHF)	exact	OMP3	-74.98104071	0.00	0.191	1.181	6.18x
O (^3P, UHF)	cholesky	OMP2	-74.96657683	0.00	0.036	0.914	25.39x
O (^3P, UHF)	cholesky	OMP3	-74.98104071	0.00	0.125	3.119	24.95x
O (^3P, UHF)	df	OMP2	-74.96656224	0.00	0.015	0.924	61.60x
O (^3P, UHF)	df	OMP3	-74.98111725	0.00	0.059	2.275	38.56x

Research impact statement

mshqc serves as the primary computational engine for electronic-structure research at the Department of Physics, Universitas Gadjah Mada, investigating $\mathcal{O}(V^4)$-memory-free orbital-optimized Møller–Plesset perturbation theory (OMP2 and OMP3) via rank-reduced tensor factorizations and spin-symmetry restoration in open-shell radicals (Hidayat, 2026). By refactoring the particle-particle ladder two-particle reduced density matrix (2-RDM) through an $\mathcal{O}(O^3V)$ intermediate (Y^{ooov}_{ijmf}) and symmetry-packed occupied-pair slicing, mshqc reduces peak molecular-orbital tensor memory in DF-OMP3 (H_2O/cc-pVQZ) by $72.32\times$ (2234.04 MB to 30.89 MB) and lowers total process Peak RSS by $1.56\times$ to $2.77\times$ compared to Psi4. In spin-contamination studies across open-shell doublet and triplet radicals (Li, C, O, OH, and biradical O_2) benchmarked against UHF and ROHF/ROMP2/ROMP3 limits, orbital relaxation in mshqc eliminates 30.13% to 83.57% of artificial spin contamination $\Delta\langle\hat{S}^2\rangle$. Furthermore, cross-auditing mshqc against Psi4 across 150 cc-pVTZ solver configurations and a 507-point cc-pVDZ validation suite (26 atomic, molecular, and ionic species; reproducible via automated GitHub Actions CI) uncovered an upstream wavefunction orbital-update bug in Psi4's dfocc module (cd_omp2_manager), where converged Cholesky-OMP2 orbitals were not propagated back to the wavefunction object prior to evaluating $\langle\hat{S}^2\rangle$. Precompiled Linux wheels on PyPI (pip install mshqc) and bundled basis sets allow researchers and students to adopt the library directly in quantum chemistry workflows.

Availability and documentation

mshqc is released under the Apache License 2.0 at https://github.com/syahrulhidayat/mshqc. Precompiled manylinux_2_28 (x86_64) wheels and source distributions are available on PyPI

DRAFT

(https://pypi.org/project/mshqc/) via `pip install mshqc`. Automated `pytest` validation suites, quickstart tutorials, and troubleshooting guides are provided in `tests/` and `docs/`.

AI usage disclosure

Generative AI tools (Gemini) were used to assist with code refactoring, profiling analysis, and manuscript editing. All code and text were validated against analytical references by the author, who assumes full responsibility for this work.

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